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Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Signature of the Supervisor: _____ Date: 01/9/2025

1. ABSTRACT

Fetal health monitoring plays a vital role in ensuring safe pregnancies and reducing complications during childbirth. However, fetal monitoring using cardiotocography (CTG) and ultrasound is typically performed only during occasional clinic visits and relies heavily on the doctor's judgment. In these critical situations, physicians welcome decision support systems that confirm and strengthen their clinical decisions. To address these challenges, this research proposes the development of an explainable, confidence-aware decision support system (DSS) for fetal health classification through a mobile application.

The proposed system leverages CTG data and employs advanced machine learning models to classify fetal health into three categories: normal, suspect, and pathological. To enhance clinical trust and usability, the system integrates an explainability module that presents model predictions in a clinically meaningful manner by aligning explanations with commonly interpreted CTG features such as baseline heart rate, variability, and accelerations/decelerations. Furthermore, a rule-based layer adapts the model outputs according to local medical guidelines, ensuring region-specific relevance and improving decision accuracy.

Unlike many existing solutions, the proposed DSS is designed to operate in a fully offline environment, enabling obstetricians and gynecologists to make informed decisions even in rural healthcare settings with limited or no internet connectivity. By combining machine learning insights, local clinical expertise, and offline capability, the system aims to assist healthcare professionals in timely detection of potential fetal risks and improve maternal and fetal outcomes.

The completed solution is expected to deliver high classification accuracy, transparent predictions, and clinically relevant recommendations, ultimately supporting safer, data-driven decision-making in diverse healthcare environments.

Keywords: Cardiotocography, Decision support system

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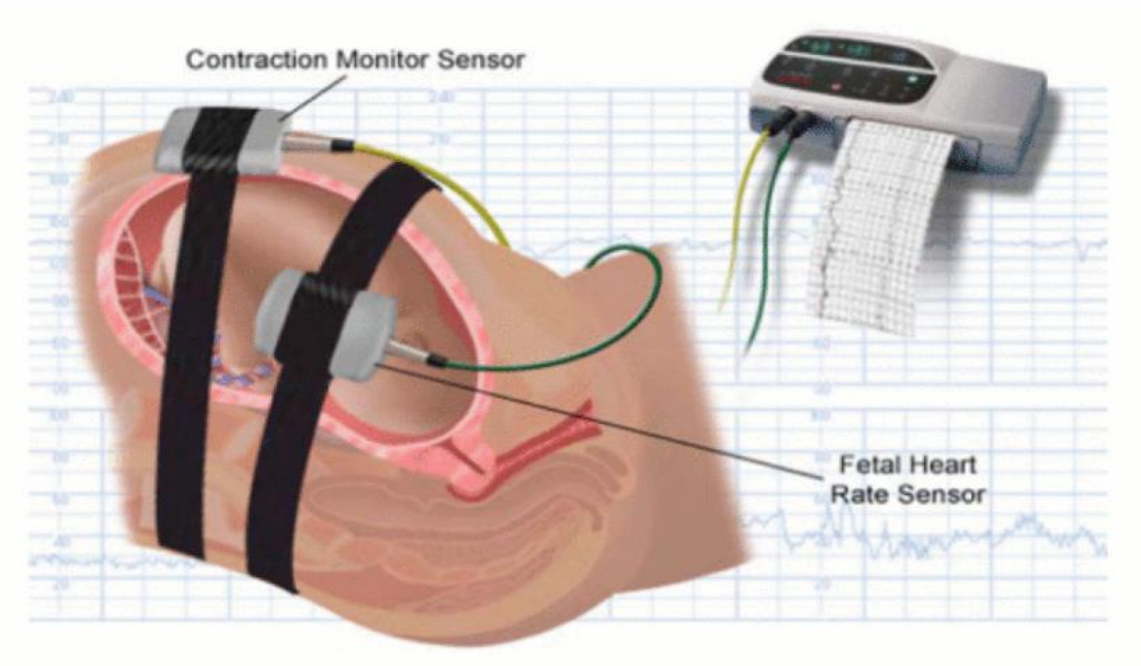
5. INTRODUCTION

5.1. Background & Literature Survey

Health complications during pregnancy remain a significant global concern, especially in developing countries where access to continuous and quality healthcare is limited. Maternal and fetal mortality rates are considerably higher in such regions due to conditions like pre-eclampsia, gestational diabetes, and inadequate fetal monitoring [1] , [2]. Fetal monitoring, normally done in the third trimester, It helps to track the well-being of the unborn child by observing fetal heart rate and uterine contractions using cardiotocography (CTG) [3]. CTG is a widely accepted, cost-effective method that enables early detection of abnormal fetal conditions, supporting clinical decision-making during pregnancy and labor [4].

Figure 2.1 A cardiotocographic (CTG) setup example



Traditional fetal health assessment relies heavily on intermittent clinical visits and subjective interpretations by healthcare professionals, which can be inconsistent and may delay timely interventions [3]. Machine learning (ML) has emerged as a promising solution to these challenges by providing automated analysis of CTG signals to classify fetal health into categories such as normal, suspicious, or pathological. Studies have demonstrated various ML techniques, including random forest, support vector machines, and neural networks, successfully improving accuracy and reliability in fetal health

classification [5], [6]. Feature selection and data preprocessing play an important role in enhancing algorithm performance for these tasks [7].

Recent developments in artificial intelligence have enabled real-time and continuous fetal monitoring systems that extend beyond well-equipped clinical settings. For example, in resource-limited areas like the Area 25 Health Centre in Malawi, AI-powered fetal assessment has contributed to an 82% reduction in stillbirths and neonatal deaths, demonstrating the life-changing impact of this technology in rural healthcare [8]. Commercial systems such as PeriWatch Vigilance provide early warning solutions combining CTG data with clinical risk factors to aid obstetric care and support timely interventions [9].

5.2. Research Gap

Most existing studies on fetal health classification using cardiotocography (CTG) data have achieved high accuracy; however, several important limitations reduce their usability in real clinical settings.

Firstly, many studies lack clinically meaningful explainability. This is an important factor missing, in fields like this research area, where understanding the reasoning behind AI decisions can directly impact patient outcomes and foster trust in the technology [16]. While some research attempt to provide feature importance or saliency maps, these explanations are more model centric and not aligned with the way obstetricians interpret CTG reports. Clinicians typically make decisions based on clinically relevant specific CTG parameters such as baseline heart rate, accelerations, decelerations, and variability patterns, but most existing models do not present explanations in this familiar context, which limits their practical value [10], [11].

Secondly, the current approaches generally do not estimate uncertainty or confidence levels for their predictions. Almost all models simply provide a class label or a raw probability score without proper calibration. In a medical context, especially for borderline or ambiguous cases, clinicians need to understand how confident the model is in its prediction to make safe decisions. The absence of uncertainty estimation reduces trust in model outputs and can lead to potential risks [12], [13].

Thirdly, existing solutions are rarely designed for offline deployment. Most models are evaluated on desktop or cloud environments and assume stable internet connectivity for inference. However, in many rural or resource-limited regions, continuous internet

access is not guaranteed. Without offline functionality, these models cannot effectively support healthcare professionals in remote clinics [12], [13].

Finally, there is no integration of local clinical guidelines in existing systems. Many countries, including Sri Lanka, follow region-specific thresholds and treatment recommendations for fetal health conditions. Current studies do not adapt their models or decision rules to these local protocols, which limits their suitability and acceptance in real-world clinical workflows [10], [14].

5.3. Research Problem

Addressing the limitations of existing fetal health classification methods introduces several research challenges.

Firstly, there is the question of how to provide clinically meaningful explanations for model predictions. Existing approaches show generic feature importance, but clinicians make decisions based on CTG-specific parameters such as baseline heart rate, accelerations, decelerations, and variability patterns. The challenge is designing explanations that align with these clinical concepts so that obstetricians and gynecologists can easily interpret and trust the system's recommendations.

Secondly, the lack of confidence scoring in current models creates a significant safety concern. In real clinical environments, especially in borderline or ambiguous cases, healthcare professionals need to understand how reliable a prediction is before acting on it. Developing a method to generate calibrated confidence levels or uncertainty scores is therefore a key research challenge.

Thirdly, enabling offline functionality is essential for supporting rural healthcare settings where internet connectivity may be limited or unavailable. The problem lies in optimizing machine learning models for on-device inference while maintaining accuracy and minimizing latency, storage, and power consumption.

Finally, adapting the system to local clinical guidelines is a challenge. Different regions follow unique protocols for interpreting CTG results and managing fetal health risks. The challenge is to integrate a rule-based layer or a guideline-aware mechanism that ensures the system's recommendations are relevant and actionable for clinicians in Sri Lanka.

6. OBJECTIVES

6.1. Main Objective

The main objective of this research is to develop a decision support system (DSS) for fetal health classification, designed to assist obstetricians and gynecologists in making accurate, transparent, and well-informed clinical decisions. By improving decision accuracy, the system aims to ensure better care for both pregnant mothers and fetuses. Furthermore, it integrates local clinical guidelines to ensure region-specific relevance and provides offline functionality to support healthcare clinicians in rural and resource-limited settings.

6.2. Specific Objectives

1. To design and develop a decision support system (DSS) capable of accurately classifying fetal health conditions using clinical data.
2. To integrate local clinical guidelines into the system to ensure that recommendations are relevant and aligned with region-specific medical practices.
3. To implement explainable AI techniques that provide transparent reasoning behind each classification, enabling clinicians to better understand the basis of decisions.
4. To incorporate confidence scoring mechanisms to indicate the reliability of each prediction, assisting healthcare professionals in making informed choices.
5. To enable offline functionality within the mobile application to ensure accessibility in rural and resource-limited settings where internet connectivity may be unavailable.

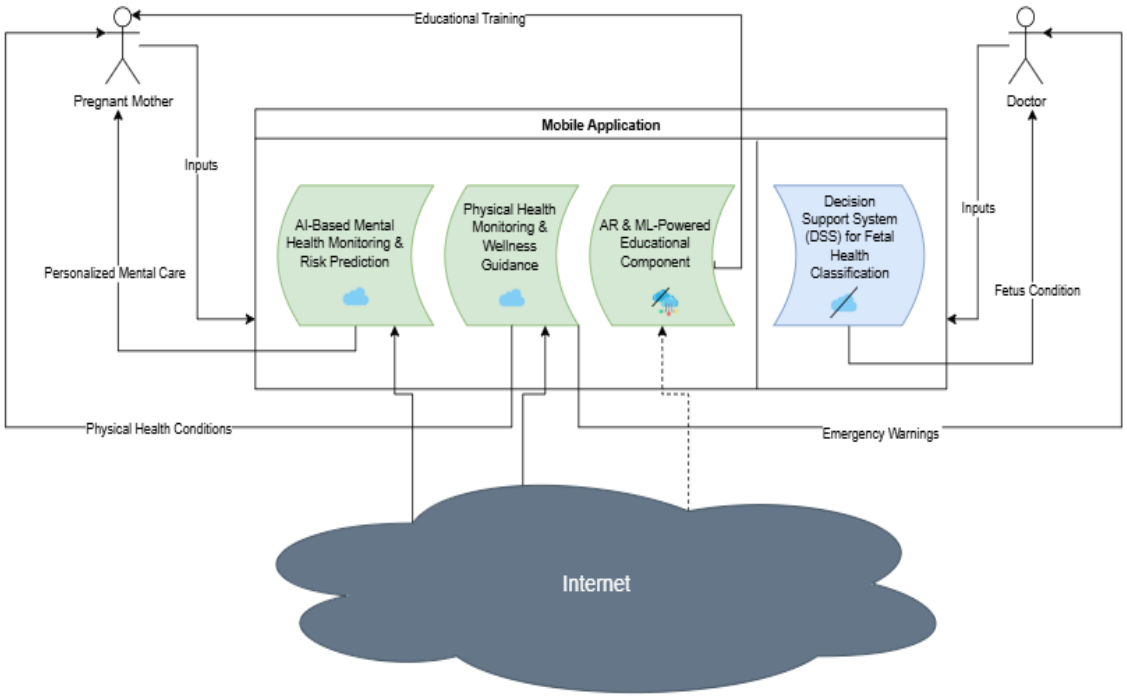
7. METHODOLOGY

The proposed project planned to complete following methodology steps. The methodology is divided into following 8 steps, which are requirement analysis, system planning and design, data preparation, model development, local medical guideline adaptation, offline capability, testing and user evaluation.

Task and sub tasks

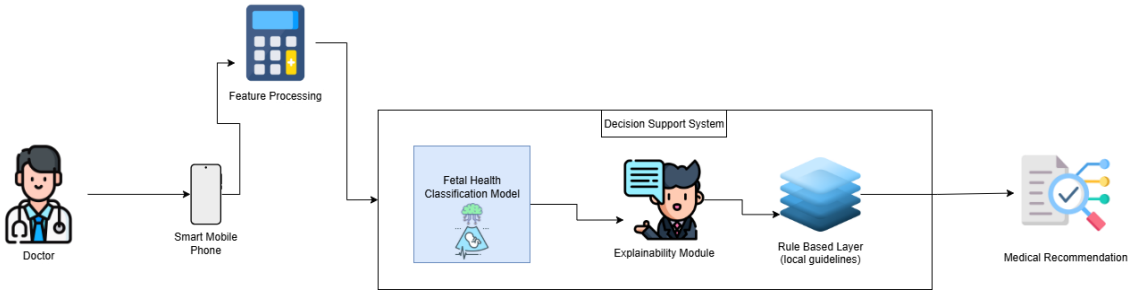
1. Requirements analysis
    - Conduct initial interviews and discussions with gynecologists and obstetrician identify specific needs and preferred methods.
    - Review the feasibility to collect CTG data
2. System planning and design
    - Outline the core objectives and features of the proposed decision support system based on previously identified research gaps.
    - Define the architecture of the mobile application, outlining the interaction flows and integration points for explainability, confidence scoring, local guideline adaptation, and offline functionality.
    - The overall system diagram is illustrated in Figure 7.2 .

Figure 7.2 Overall system diagram



- The individual system diagram is illustrated in Figure 7.3 .

Figure 7.3 Individual system diagram



3. Data preparation

- Data cleaning – Checked for missing values, duplicates, and inconsistencies to ensure dataset quality.

- Data analysis – Analyzed feature distributions, correlations, and class imbalance.
- Data sampling – Applied techniques like oversampling to handle class imbalance in fetal health categories.
- Feature scaling – Standardized feature values to improve model performance and training stability.
- Dataset splitting – Divided the dataset into training and testing subsets for unbiased model evaluation.

4. Model development

- Use the fetal health classification datasets to train models such as Random Forest and XGBoost.
- Develop an ensemble model to improve prediction accuracy.
- According to the ScienceDirect paper [17], a broad range of interpretability methods is available for explainable AI in healthcare. These methods can be categorized into several main families, as discussed in the ArXiv survey [18]. Among these, game-theory-based methods are particularly relevant to this research component because the proposed system heavily relies on feature-driven predictions. In these methods, each feature is treated as a player in a cooperative game, and the algorithm calculates the contribution of each feature to the final decision. Furthermore, game-theory-based techniques, such as SHAP, provide both local and global interpretability, allowing clinicians to understand individual predictions while also gaining insights into the overall model behavior [17]. For these reasons, this research adopts a game-theory-based interpretability approach.
- Add confidence scoring to enhance clinical trust and highlight uncertain predictions.
- Convert the model to mobile friendly format.

5. Local medical guideline adaptation

- Consult with Sri Lankan clinicians to understand local thresholds and response practices.

- Implement a rule-based layer on top of ML predictions to adapt outputs to local healthcare needs.

6. Offline capability

   - Build an offline-capable application that accepts CTG derived variables and outputs fetal health classification. Ensure usability in rural settings with limited or no internet access.

7. Testing

   - Conduct unit testing of individual components and system-level testing for latency, accuracy, and power efficiency.

   - Perform accessibility compliance checks and optimize for mobile hardware constraints.

8. User evaluation

   - Conduct user evaluation by handing over the application to clinicians working in this healthcare sector, allowing them to test the system in real world clinical scenarios.
   - Collect quantitative measures (accuracy, processing time) and qualitative feedback (user satisfaction, ease of use).

Materials and tools

- Equipment:

  - Android smartphone.

  - Laptop/PC for model training and development.

- Software & Frameworks:

  - Python and TensorFlow.

  - Google Colab for model training.

  - Android Studio and VS code for mobile app development.

- Other Resources:

- Fetal health classified dataset

Data collection

- Fetal health classified dataset : Datasets are available in Kaggle.

Anticipated conclusion

The completed system will provide obstetricians and gynecologists with an explainable**,** confidence-aware decision support system for fetal health classification through a mobile application. It is expected to deliver high classification accuracy, transparent predictions with explainability, and region specific recommendations by integrating local medical guidelines. With its offline capability, the system will remain accessible in rural and resource-limited healthcare settings, enabling timely and informed clinical decisions. The final solution aims to improve maternal and fetal care outcomes while supporting healthcare professionals in making safe, data-driven decisions.

8. PROJECT REQUIRMENTS

8.1. Functional Requirements

These are the essential features for the Fetal Health Classification. They focus on helping clinicians through different aspect to ease the clinicians decision making process to classify fetal health.

1. Fetal Health Classification – The system needs to be able to classify fetal health based on fetus CTG result accurately.
2. Explainable Decision – System needs to be able to clearly explain the decision It made based on which input features and which feature impact most and least to made that decision.
3. Confidence Decision – Decisions system made suppose to have a confidence score to validate the decision.
4. Local clinical guidelines – Decisions system made needs to follow local specific medical practices.

8.2 Non Functional Requirements

1. Offline Capability – Machine learning model needs to bundled within the system itself, allowing the app to work seamlessly even without an internet connection.

2. Privacy & Security - User data is supposed to processed and stored locally on the device, minimizing the need to send sensitive information to external servers. This is a crucial requirement for industries such as healthcare.

3. Lower Operational Cost – System needs to be able to maintain without any third party service providers for cloud infrastructure, data storage and bandwidth.

4. Efficiency – System needs be optimized to use user device resources efficiently, leading to lower power consumption.

5. Reliability - System needs to be able to run continually even when network connectivity slow or unavailable.

9. BUDGET AND BUDGET JUSTIFICATION

Table 7.1 : Budget

Description	Occurrence	Cost
Mobile App Hosting	One-Time	LKR 7500
Model Training	Pay-as-you-go	LKR 3000 for 100 GPU units

1. Mobile App Hosting – One-Time – LKR 7,500
   A reliable hosting service is required to deploy the mobile application and ensure seamless accessibility for healthcare practitioners. This cost covers a one-time setup for hosting services, which will support system updates, secure storage, and stable application performance.

2. Model Training – Pay-as-you-go – LKR 3,000 for 100 GPU units
   Training the machine learning models for fetal health classification requires high-performance computing resources. A GPU-enabled cloud service will be utilized to efficiently train and fine-tune the models. The estimated cost of LKR 3,000 for 100 GPU units ensures faster experimentation, better model accuracy, and reduced development time.

10. FAILURE MODE ANALYSIS

Table 10.2 : Failures which might happen

Failure Name	Effect	Mitigation action
Confidence score	Clinicians won't know the trustability of the outcome.	Research about the new techniques to achieve reliable outcome.
Hosting model on mobile	Won't be able to deliver one mobile app for all 4 components.	Will have to create a desktop application only for my component separate from other system components.
Rule based layer for local health guidelines	Model won't adapt to local guidelines	Get the expert clinician help to develop this rule based layer.

11. GANTT CHART

Table 11.3: Gantt chart

Tasks	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May
Requirement Analysis									
System Planning & Design									
Data Finding									
Data Preparation									
Feature Processing									
Model Development									
Explainability Module									
Local Medical Guideline Adaptation									
Offline System Packaging									
Convert Model to Mobile ML format									
Build a Mobile Application									
Testing									
User Evaluation									
Final Report & Documentation									

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